

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Industry Problem solving task
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A high-frequency financial trading system must process large volumes of market data with extremely low response time. Analyze the roles of CPU caches, DRAM, and persistent storage in handling frequently accessed trading data. Considering latency, data locality, capacity, and bandwidth, propose a memory hierarchy that minimizes data-access delays and supports high transaction throughput.
2. A medical imaging workstation processes large CT and MRI datasets while simultaneously running image-enhancement and visualization applications. Evaluate the interaction between cache memory, DRAM, and high-capacity storage during image processing. Propose an optimized memory hierarchy that balances access latency, memory capacity, bandwidth, and cost to improve overall processing performance.
3. A smart surveillance system receives continuous video streams from multiple high-resolution cameras and must detect objects in real time. Analyze the memory requirements for storing video frames, intermediate processing data, and frequently accessed detection parameters. Compare cache, DRAM, and non-volatile memory technologies and design a hierarchy that minimizes memory bottlenecks while maintaining high data throughput.
4. A scientific computing system performs large-scale simulations involving matrices and continuously accessed numerical datasets. Analyze how cache capacity, data locality, cache misses, DRAM bandwidth, and memory latency affect the execution of computational kernels. Recommend memory-hierarchy improvements that can reduce cache misses, increase effective memory bandwidth, and improve overall simulation performance.
5. A wearable health-monitoring device continuously collects sensor readings while operating under strict battery constraints. Analyze the trade-offs among SRAM, DRAM, and non-volatile memory in terms of latency, capacity, energy consumption, and data retention. Design a memory hierarchy that provides rapid access to frequently processed sensor data while minimizing power consumption and extending battery life.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professionalism(5)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10		/ 100

Course Coordinator

Faculty Incharge